Practical 4

Ex.1 Standard Normal Distribution

*Find the probability	z	Φ(z)	Equation	Answer
(a) Between 0.87 and 1.28	0.87	0.8078	$P(z_1<Z<z_2) = \Phi(z_2) - \Phi(z_1)$	**0.0919**
	1.28	0.8997		
(b) Between −0.34 and 0.62	-0.34	0.3669	$P(z_1<Z<z_2) = \Phi(z_2) - \Phi(z_1)$	**0.3654**
	0.62	0.7324		
(c) Greater than 0.85	0.85	0.8023	$P(Z>z) = 1 - \Phi(z)$	**0.1977**
(d) Greater than −0.65	-0.65	0.2578	$P(Z>z) = 1 - \Phi(z)$	**0.7422**

Ex.2 Driver Reaction Time

Items	Value
Mean reaction time, μ (sec)	1.25
Standard deviation, σ (sec)	0.46

*Find the probability	x	z	Equation	Answer	Check (NORM.DIST)
(a) Between 1.00 sec and 1.75 sec	1.00	-0.5435	$P(x_1<X<x_2) = \Phi(z_2) - \Phi(z_1)$	**0.5681**	0.5681
	1.75	1.0870			
(b) Reaction time exceeds 2 sec	2.00	1.6304	$P(X>x) = 1 - \Phi(z)$, $z = (x - \mu)/\sigma$	**0.0515**	0.0515

Ex.3 Current Measurement in a Strip of Wire

Items	Value
Mean, μ (milliamperes)	10
Variance, σ^2 (milliamperes^2)	4
Standard deviation, σ = √variance	2.00

*Find the probability	x	z	Equation	Answer	Check (NORM.DIST)
(a) Measurement exceeds 13 mA	13.00	1.5000	$P(X>x) = 1 - \Phi(z)$, $z = (x - \mu)/\sigma$	**0.0668**	0.0668
(b) Between 9 mA and 11 mA	9.00	-0.5000	$P(x_1<X<x_2) = \Phi(z_2) - \Phi(z_1)$	**0.3829**	0.3829
	11.00	0.5000			

Ex.4 Finding x for a Given Probability

Items	Value
Mean, μ	54
Standard deviation, σ	8
(a) Given P(X < x)	0.80
(b) Given P(X > x)	0.10

*Find the value of x	Area to left	z	Equation	Answer (x)	Check (NORM.INV)
(a) x such that P(X < x) = 0.80	0.8000	0.8416	$z = \Phi^{-1}$(area to left); $x = \mu + z\cdot\sigma$	**60.7330**	60.7330
(b) x such that P(X > x) = 0.10	0.9000	1.2816	area to left = 1 − P(X>x); $x = \mu + z\cdot\sigma$	**64.2524**	64.2524

Ex.5 Vehicle Speed on a Bridge

Items	Value
Speed x_1 (m/h)	39.12
$P(X < x_1)$	0.05
Speed x_2 (m/h)	73.24
$P(X > x_2)$	0.1

*(a) Find the mean and SD	Symbol	value	Equation	Answer
[1] z-value for the lower 5% point	z_1	0.0500	$z_1 = \Phi^{-1}(P(X<x_1))$	**-1.6449**
[2] z-value for the upper 10% point	z_2	0.9000	$z_2 = \Phi^{-1}(1 - P(X>x_2))$	**1.2816**
[3] Standard deviation	σ	34.1200	$\sigma = (x_2 - x_1) / (z_2 - z_1)$	**11.6594**
[4] Mean	μ	39.1200	$\mu = x_1 - z_1\cdot\sigma$	**58.2979**

*Find the probability	x	z	Equation	Answer	Check (NORM.DIST)
(b) Speed between 50 and 65 m/h	50.00	-0.7117	$P(x_1<X<x_2) = \Phi(z_2) - \Phi(z_1)$	**0.4790**	0.4790
	65.00	0.5748			
(c) Speed exceeds the 70 m/h limit	70.00	1.0037	$P(X>x) = 1 - \Phi(z)$, $z = (x - \mu)/\sigma$	**0.1578**	0.1578